

Ultraefficient Civil Transport

The 21st century brought even greater awareness of the need to economize on fossil fuels. One solution was the largest passenger airliner yet, with room for over 850 passengers; another was to use composite construction for light weight and durability for both airliners and business jets that can legally be operated with one pilot (cutting costs and increasing capacity).

▷ **Piper PA-46-500TP Malibu Meridian 2000**
Origin US
Engine 500 hp Pratt & Whitney Canada PT6A-42A turboprop
Top speed 301 mph (484 km/h)

Certificated in 2000, the turboprop version of Piper's 1979 piston-engined Malibu, the Meridian, was given larger wing and tail surfaces and a new instrument panel; it sold in 2012 for \$2.13 million.



◁ **Beechcraft Premier I 2001**
Origin US
Engine 2 x 2,300 lb (1,043 kg) thrust Williams International FJ44-2A turbofan
Top speed 530 mph (853 km/h)

With a spacious and comfortable cabin, the Premier's carbon-fiber/epoxy honeycomb fuselage construction made it light enough to be operated by a single pilot and capable of record speeds.



△ **Bombardier Learjet 45XR 2004**
Origin Canada/US
Engine 2 x 3,500 lb (1,588 kg) thrust Honeywell TFE731-20BR turbofan
Top speed 535 mph (861 km/h)

Bombardier upgraded the successful midsize 1990s Learjet 45 to XR specification in 2004, giving it more powerful engines and offering faster climb rate, higher weight capacity, and increased cruise speed.

△ **Dassault Falcon 900C 2000**
Origin France
Engine 3 x 4,750 lb (2,155 kg) thrust AlliedSignal TFE731-5BR-1C turbofan
Top speed 590 mph (950 km/h)

Dassault's intercontinental business jet for the 21st century boasted three engines, giving excellent performance as well as increased range, enhanced by the latest Honeywell avionics.

▷ **Gulfstream G150 2002**
Origin US
Engine 2 x 4,420 lb (2,005 kg) thrust Honeywell TFE731-40AR turbofan
Top speed 631 mph (1,015 km/h)

Gulfstream reentered the midsize market, giving the G150 a near-square cabin cross section with more useful space than circular cabins, and major improvement on all fronts over the G100.



△ **Airbus A380 2005**
Origin European Consortium
Engine 4 x 84,000 lb (38,102 kg) thrust Rolls-Royce Trent 900 or 81,500 lb (36,968 kg) thrust Engine Alliance GP7000 turbofan
Top speed 587 mph (945 km/h)

The world's largest passenger airliner of the early 21st century, the double-deck A380 required airports to enlarge their facilities to cope with it. Singapore Airlines flew the first one in 2007.

◁ **Gulfstream G550 2003**
Origin US
Engine 2 x 15,385 lb (6,978 kg) thrust Rolls-Royce BR710 turbofan
Top speed 585 mph (941 km/h)

With range and performance increased over the Gulfstream V by reducing aerodynamic drag, the 550 has the longest range in its class at 7,767 miles (12,500 km). It also has Enhance Vision for landing in fog.

